MUTUAL FUND PERFORMANCE & SIP ANALYTICS DASHBOARD

Final Project Report

Fintech Data Analytics Internship

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Abstract

The Mutual Fund Performance & SIP Analytics Dashboard project was developed to analyze mutual fund industry trends using historical NAV data, SIP inflows, category-wise investments, and fund performance metrics. The project demonstrates the complete data analytics workflow including data collection, data cleaning, exploratory data analysis (EDA), visualization, dashboard creation, and reporting. Using Python and Power BI, raw financial datasets were transformed into actionable insights that help understand investor behavior, fund growth patterns, and category performance.

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1. Introduction

The mutual fund industry has witnessed significant growth due to increasing investor participation and the popularity of Systematic Investment Plans (SIPs). Financial institutions and investors require analytical tools to monitor fund performance, track investment trends, and evaluate growth opportunities.

This project aims to build an interactive analytics solution that enables users to analyze mutual fund performance through meaningful visualizations and business insights.

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2. Project Objectives

The primary objectives of this project are:

- Analyze historical Net Asset Value (NAV) trends.
- Study SIP inflow growth over time.
- Examine category-wise investment patterns.
- Identify top-performing mutual fund categories.
- Build an interactive Power BI dashboard.
- Generate business insights from financial data.

- Present findings through reports and visualizations.

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3. Dataset Description

The project uses multiple datasets related to mutual funds and SIP investments.

Key Datasets

1. NAV Historical Data
    - Daily NAV values
    - Scheme-wise performance tracking
2. SIP Inflows Data
    - Monthly SIP contributions
    - Active SIP accounts
    - New SIP registrations
3. Category Inflows Data
    - Equity category inflows
    - Debt category inflows
    - Hybrid category inflows
4. Scheme Information Data
    - Scheme names
    - Fund categories
    - Fund houses
5. AUM Data
    - Assets Under Management
    - Category-wise asset distribution

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4. Technology Stack

The following tools and technologies were used during the project:

Programming

- Python

- SQL

Libraries

- Pandas
- NumPy
- Matplotlib
- Plotly

Data Visualization

- Power BI

Version Control

- Git
- GitHub

Development Environment

- Jupyter Notebook
- VS Code

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5. Data Cleaning and Preprocessing

Raw financial data often contains inconsistencies and missing values. Several preprocessing steps were performed to ensure data quality.

Cleaning Activities

- Removed duplicate records.
- Handled missing values.
- Converted date columns into datetime format.
- Standardized column names.
- Validated data types.
- Corrected formatting inconsistencies.
- Prepared datasets for analysis and dashboard integration.

Outcome

The cleaned datasets provided accurate and reliable information for further analysis and visualization.

6. Exploratory Data Analysis (EDA)

EDA was conducted to understand trends, relationships, and patterns within the data.

NAV Trend Analysis

Historical NAV values were analyzed to observe scheme growth over time.

Observations

- Most schemes displayed long-term upward movement.
- Short-term fluctuations reflected market volatility.
- Consistent NAV growth indicated stable fund performance.

SIP Inflow Analysis

Monthly SIP investments were examined to evaluate investor participation.

Observations

- SIP contributions showed a positive growth trend.
- Active SIP accounts increased steadily.
- Investor confidence remained strong across periods.

Category-wise Inflow Analysis

Investment inflows were analyzed across different fund categories.

Observations

- Equity-oriented funds attracted the highest investments.
- Hybrid funds showed moderate growth.
- Debt funds demonstrated stable investment behavior.

Performance Comparison

Multiple schemes were compared using available performance indicators.

Observations

- Certain schemes consistently outperformed peers.
- Growth-oriented categories generated higher returns.
- Performance differences highlighted category preferences.

7. Dashboard Development

An interactive Power BI dashboard was created to provide a comprehensive overview of mutual fund performance.

Dashboard Features

KPI Cards

- Total SIP Inflows
- Total Assets Under Management (AUM)
- Average NAV
- Active SIP Accounts

Visualizations

- Monthly SIP Trend
- Category-wise Inflows
- NAV Growth Trend
- Top Performing Categories
- Fund Performance Comparison

Interactive Components

- Date Filters
- Category Slicers
- Drill-down Functionality
- Dynamic Visual Updates

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8. Key Insights

Several important business insights were generated from the analysis.

Investor Behavior

- SIP participation continues to increase.
- Long-term investment strategies remain popular.

Category Performance

- Equity funds attract higher investment volumes.

- Investors prefer growth-oriented investment products.

Fund Growth

- Strong NAV growth indicates positive market participation.
- Consistent inflows support long-term asset expansion.

Business Opportunities

- Financial institutions can promote SIP products further.
- Data-driven monitoring can improve investment decision-making.

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9. Challenges Faced

During the project, several challenges were encountered:

- Data inconsistencies across datasets.
- Missing values in financial records.
- Standardizing formats from multiple sources.
- Building efficient dashboard relationships.
- Ensuring accurate analytical calculations.

These challenges were resolved through systematic cleaning and validation processes.

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10. Conclusion

The Mutual Fund Performance & SIP Analytics Dashboard successfully transformed raw financial datasets into meaningful insights. Through data cleaning, exploratory analysis, and dashboard development, the project provided a comprehensive view of mutual fund industry trends.

The final solution enables users to analyze NAV movements, monitor SIP growth, compare fund categories, and understand investment behavior through interactive visualizations. The project demonstrates practical application of Python, SQL, Power BI, and data analytics techniques in the financial domain.

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11. Future Scope

Potential future enhancements include:

- Real-time NAV integration through APIs.

- Predictive analytics for fund performance forecasting.
- Risk-adjusted return analysis.
- Portfolio recommendation systems.
- Machine learning models for investment insights.
- Automated reporting and alerts.

References

- Mutual Fund Industry Reports
- AMFI Data Sources
- Financial Market Publications
- Python Documentation
- Power BI Documentation

Acknowledgement

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Thank You.